

1. The purpose of a loop structure is to execute lines of code repeatedly until the conditions listed are met. It reduces code duplication and makes the code cleaner.
2. The difference between a while loop and a do-while loop is that a while loop checks the condition before the loop begins. So it would error if false, but a do-while loop will run at least once no matter what.
3. We did use this in NumbersSum Skillbuilder.
4. a) When the code executes, it repeatedly loops because there are no conditions written for it to stop; it loops forever.
b) If conditions are not written properly, or if existing conditions are not defined.
c) If it is an infinite loop, the computer keeps remembering each loop; however, the memory will eventually run out, as it can only store so much. This leads to overflow, which is an error message saying memory capacity has been exceeded.
5. 60 times.
6. Any value less than 120.
7. A counter counts how many times something happens; however, an accumulator tracks a running total of values. A counter is used to limit and count how many loops are done so you do not cause overflow. It also tracks events in code. Accumulators calculate financial totals and averages.
8.

```
int sum = 0;
for (int i = 3; i <= 10; i++) {
    sum += i;
}
```
9. You chose based on whether you understand the amount or iterations before running the code or after running the program.
11. a) 9
b) my
c) my string
d) my STRING
e) my string